



上海交通大学学位论文

面向 3C 工业生产线的双臂机器人视觉-语言-动作模型

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DUAL-ARM ROBOTS IN 3C INDUSTRIAL

PRODUCTION LINES

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摘 要

现代 3C 电子装配线需要处理小型元器件、高产品多样性和富接触操作，这些任务仍难以通过传统固定程序机器人实现自动化。本项目针对这一缺口，面向双臂工业操作开发视觉-语言-动作 (Vision-Language-Action, VLA) 系统。系统设计运行于双臂 Franka 平台，并面向典型 PC 关键部件任务，包括内存条插入、CPU 放置以及电源线连接。

当前部署系统选择相应的 ACT 检查点，接收同步视觉观测和机器人状态，并在闭环执行过程中输出机器人及夹爪动作。现阶段证据限于干运行和有边界的验证活动；本文不声称已经实现语言条件推理、统一多任务控制、自主规划、故障检测、恢复或双臂协同操作。这些功能仍是面向预期 VLA 与双臂系统的长期扩展方向。

工程验收目标包括：规划成功率高于 90%，任务执行成功率高于 70%，部件检测精度高于 90%，位姿误差低于 20 mm，故障检测率高于 90%，恢复成功率高于 70%。这些数值定义验证阈值，而非已经证明的性能结果。

关键词：双臂机器人，视觉-语言-动作模型，工业自动化

ABSTRACT

Modern 3C electronics assembly lines deal with small components, high product variety, and contact-rich operations that remain difficult to automate with conventional fixed-program robots. Our project targets this gap by developing a Vision-Language-Action (VLA) system for bimanual industrial manipulation. The system is designed to run on a dual-arm Franka platform and to handle representative PC key-component tasks—specifically RAM insertion, CPU placement, and power-cable connection.

The deployed system selects the corresponding ACT checkpoint, receives synchronized visual observations and robot state, and outputs robot and gripper actions during closed-loop execution. Current evidence is limited to dry runs and bounded validation activities; the report does not claim demonstrated language-conditioned inference, unified multi-task control, autonomous planning, failure detection, recovery, or bimanual manipulation. These functions remain long-term extensions toward the intended Vision-Language-Action (VLA) and dual-arm system.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results.

Key words: dual-arm robot, vision-language-action model, industrial automation

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Chapter 1 Introduction

1.1 Background and significance

The 3C industry—computer, communication, and consumer electronics—increasingly demands flexible robotic systems that can cope with short product life cycles, diverse part geometries, small components, and frequent line changeovers. Traditional hard-coded automation approaches are effective for stable, high-volume production, but they exhibit significant limitations in high-mix assembly scenarios. Each product variant typically requires reprogramming, fixture redesign, and system reconfiguration, which increases development and maintenance costs. In contrast, human operators can rapidly understand new tasks from natural-language instructions and adapt to new operating conditions with limited prior programming. This contrast highlights the importance of advancing industrial robotic systems toward more human-aligned capabilities, especially semantic understanding of instructions and data-efficient adaptation under limited demonstrations.

Recent advances in Vision-Language-Action (VLA) models offer a promising route toward this goal. By jointly learning from visual observations, language instructions, robot proprioceptive states, and action data, VLA models can in principle generalize across tasks without requiring separate hand-engineered programs for every product variant. Landmark systems, including RT-2, OpenVLA, RDT-1B, and the π series, demonstrate the potential of transferring semantic knowledge from large vision-language models to physical robot control^[1-4]. Earlier work on visual and language-conditioned manipulation also established complementary design patterns: Transporter Networks use spatially grounded pick-and-place representations, CLIPort combines semantic and spatial pathways, and PerAct couples language goals to structured visual observations^[5-7]. In parallel, BridgeData V2, Open X-Embodiment, DROID, and LIBERO have made dataset scale, embodiment diversity, and reproducible task evaluation central considerations in robot-learning system design^[8-11].

Despite this progress, adapting VLA-style systems to industrial assembly remains a system-level engineering problem. Many published demonstrations emphasize tabletop manipulation, single-arm control, or short-horizon tasks with relatively forgiving tolerances. In contrast, 3C assembly requires reliable perception of small components, accurate contact-rich

manipulation, synchronized sensing, safe robot execution, and repeatable validation under bounded physical conditions. These requirements motivate a complete manipulation pipeline that links hardware selection, teleoperation, data collection, model training, deployment, and quantitative evaluation.

1.2 Design problem

This project aims to prototype a dual-arm industrial manipulation platform that couples camera observations and robot proprioceptive state to learned task actions for a scoped set of PC assembly operations. The target workcell consists of two Franka Research 3 (FR3) robot arms with complementary global and wrist/side camera views. Demonstrations are collected through a GELLO-based teleoperation interface and are recorded as synchronized visual observations, robot states, and robot or gripper actions. The intended task domain covers representative PC key-component operations, including RAM insertion, CPU placement, and power-cable connection.

At the present stage, task-specific Action Chunking with Transformers (ACT) checkpoints are selected by the operator rather than inferred from a language command^[12]. The deployed system receives synchronized observations and robot state, then outputs robot and gripper actions for closed-loop execution. This implementation is therefore best understood as a staged imitation-learning baseline and a foundation for the intended VLA system, rather than as a fully demonstrated language-conditioned, unified multi-task controller. Language-conditioned inference, autonomous planning, failure detection, recovery, and robust bimanual coordination remain long-term extensions toward the final VLA and dual-arm manipulation concept.

1.3 Project objectives and scope

The project objective is to establish an end-to-end engineering pipeline for bounded industrial manipulation experiments. This pipeline includes hardware integration, synchronized multi-camera sensing, teleoperated demonstration collection, dataset preparation, ACT-based policy training, policy deployment, and repeatable physical validation. The expected Design Expo outcome is a measurable real-robot demonstration of the scoped assembly step,

including task success, pose-estimation performance, and failure-handling evaluation.

The engineering acceptance targets are planning success above 90%, task execution success above 70%, component detection accuracy above 90%, pose error below 20 mm, failure-detection rate above 90%, and recovery success above 70%. These values define validation thresholds rather than demonstrated performance results. Current evidence is limited to dry runs and bounded validation activities; therefore, this thesis does not claim unrestricted open-world generalization, autonomous factory deployment, guaranteed safe recovery from every failure mode, or fully demonstrated VLA-based multi-task inference.

1.4 Main contributions

The main contributions of this thesis are as follows:

1. A dual-FR3 manipulation platform is specified and integrated with GELLO teleoperation, synchronized camera sensing, robot-state recording, and task-compatible end effectors and fixtures.
2. A reproducible data and learning pipeline is developed to convert teleoperated demonstrations into datasets for ACT-based training and real-robot deployment.
3. A scoped industrial assembly validation framework is defined for representative PC component tasks, with quantitative targets for planning, execution, perception, pose accuracy, failure detection, and recovery.
4. A traceable engineering process links customer requirements and quality function deployment to concept generation, module selection, implementation decisions, and validation criteria.

1.5 Organization of this thesis

The remainder of this thesis is organized as follows. Chapter 2 translates customer requirements into measurable engineering specifications and presents the quality function deployment. Chapter 3 describes functional decomposition, concept generation, and selection of the major hardware, sensing, teleoperation, end-effector, and policy modules. Chapter 4 details the final system design, data and control architecture, model-training procedure, and implementation plan. Chapter 5 presents the physical validation methods and experimen-

tal results. Chapter 6 discusses the strengths and limitations of the system, summarizes the conclusions, and identifies directions for future work.

Chapter 2 Design specification

2.1 Customer requirements

Through discussions with the project sponsor and analysis of the target application, eight customer requirements were identified and assigned importance weights reflecting the sponsor's priorities. The requirements capture both functional goals and non-functional concerns, including robustness, safety, generalizability, cost, and reliability.

Table 2–1 Customer requirements and importance weights

No.	Customer Requirement	Weight
CR1	Complete assembly task successfully	0.18
CR2	High assembly precision	0.16
CR3	Robust to environment and disturbances	0.15
CR4	Safe and collision-free operation	0.14
CR5	Fast task execution	0.12
CR6	Easy to use and generalize	0.10
CR7	Cost-effective system	0.08
CR8	System reliable and recovery capable	0.07

The customer requirements were translated into engineering specifications through the QFD process rather than by assigning targets independently. For each customer requirement i and engineering specification j , the team assigned a relationship value $r_{ij} \in \{0, 1, 3, 9\}$ (none, weak, moderate, or strong) and calculated the unnormalized engineering priority as

$$I_j = \sum_i w_i r_{ij}, \quad (2-1)$$

where w_i is the customer-importance weight in Table 2–1. The priorities were then normalized to identify the technical characteristics that most directly protect the high-weight needs. CR1 and CR5 primarily drive ES1 and ES2; CR2 drives ES3 and ES4; CR3, CR4, and CR8 drive ES5 and ES6; and CR6–CR7 constrain the selected implementation to a usable, feasible, and maintainable route. This translation produces an auditable chain from the sponsor's language to acceptance tests and explains why the later concept search is organized around platform, teleoperation, data collection, gripper, and policy modules.

2.2 Engineering specifications

The customer requirements were converted into six measurable engineering specifications. The targets in Table 2–2 are the final acceptance thresholds for the present proof-of-concept, rather than performance claims. Each rate is calculated over independently initialized physical trials of the relevant task. A trial is counted once only; an unsuccessful recovery is not reclassified as an independent execution attempt. This convention prevents recovery behavior from artificially inflating the reported task-execution success rate.

Table 2–2 Final engineering specifications and acceptance thresholds

No.	Engineering Specification	Target	Operational Definition
ES1	Planning success rate	> 90%	Fraction of trials in which the selected task policy and initialized scene form a valid executable plan.
ES2	Task execution success rate	> 70%	Fraction of trials in which the prescribed assembly operation reaches its task-specific completion condition.
ES3	Component detection accuracy	> 90%	Fraction of evaluated observations in which the required component is correctly detected for the active task.
ES4	Pose estimation error	< 20 mm	Euclidean position error between the estimated and reference component pose.
ES5	Failure detection rate	> 90%	Fraction of induced or naturally occurring execution failures that are identified before further unsafe action is issued.
ES6	Recovery rate	> 70%	Fraction of detected failures for which the defined recovery procedure restores a valid task state.

2.3 Quality function deployment

Figure 2–1 records the resulting House of Quality. Its 9–3–1 relationship scale makes the engineering priorities interpretable rather than decorative: task completion, component localization, pose estimation, and failure handling carry influence across multiple high-weight customer requirements. The roof is used as a design-control device. For example, additional visual coverage can strengthen ES3 and ES4, but it increases calibration, synchronization, and data-management burden; similarly, recovery logic can improve ES5–ES6 while adding

controller integration work. These QFD trade-offs define the subsequent concept space. The platform and gripper must enable precise contact motion for ES1–ES2 and ES4; the tele-operator and data pipeline must produce usable demonstrations and observable scenes; and the policy must convert those observations into safe executable actions while leaving explicit hooks for failure monitoring and recovery.

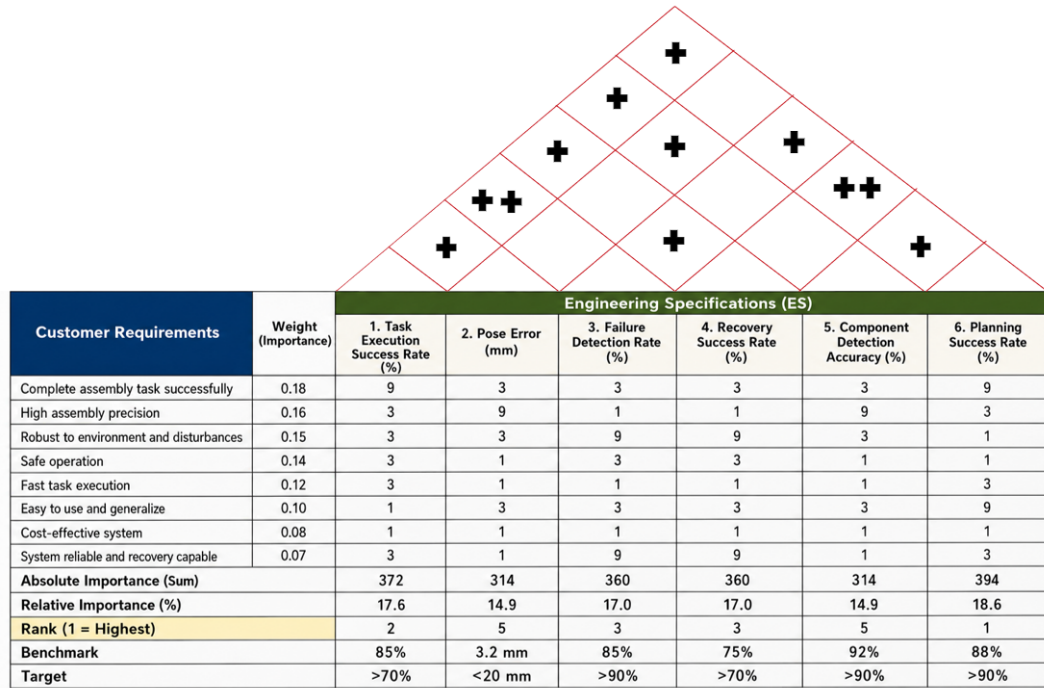


Illustration 2–1 QFD House of Quality for the VLA bimanual manipulation system.

Chapter 3 Concept generation and selection

3.1 Concept generation

3.1.1 Functional decomposition

The target assembly system was first decomposed into solution-neutral functions: establish the workspace state; create a stable robot–operator interface for demonstrations; record synchronized visual, proprioceptive, and action data; grasp and constrain the component during contact; map the current observation to coordinated motion; and detect or recover from an invalid execution state. The decomposition converts the QFD priorities into five design modules. The robot platform provides repeatable bimanual motion for ES1, ES2, and ES4; the teleoperation module enables usable demonstrations; the data-collection module supports ES3–ES4 and learning reproducibility; the gripper realizes component contact; and the policy maps observations and state to actions. Failure monitoring and recovery remain cross-cutting functions because they act across the collection and deployment loop.

3.1.2 Structured brainstorming

The team used structured brainstorming to generate alternatives for each module before choosing a configuration. Candidate ideas were retained when they addressed at least one QFD-linked specification and did not violate laboratory safety, hardware access, or the project schedule. The session considered the available FR3 ecosystem alongside Flexiv and ALOHA-style platforms; direct joint-master, XR, and hand-held teleoperation; progressively richer observation schemas; compliant and rigid end-effectors; and policy families ranging from Diffusion Policy to ACT and future VLA models. This procedure deliberately separates the choice of a data-collection interface from the choice of a learning algorithm, since neither can be justified independently of the sensing and robot-control path.

The generated alternatives were also informed by established research directions rather than by ad hoc feature lists. Robomimic and MimicGen motivate treating demonstration quality and dataset construction as system-level design variables^[13-14]; RVT, 3D Diffusion Policy, and R3M respectively motivate the retained alternatives for multi-view geometry, 3D visuomotor control, and reusable robot visual representations^[15-17]. These references

define the external concept space, while the selected configuration remains constrained by the laboratory hardware, the QFD priorities, and the project schedule.

3.1.3 Morphological chart and generated module concepts

Table 3–1 records the resulting concept space. A row represents one functional module, and a column entry represents one candidate realization. The bold entry in each row is the selected module concept; the table is not a claim that the unselected entries were all built or experimentally benchmarked.

Table 3–1 Morphological chart for the five functional modules

Concept	Candidate A	Candidate B	Candidate C
C1: Platform	Dual Franka FR3	Dual Flexiv platform	ALOHA-style bimanual platform
C2: Teleoperator	GELLO joint-master	PICO 4 Ultra XR	UMI hand-held
C3: Data collection	Two wrist + one top camera + robot state/action	One wrist + one top camera + robot state/action	Robot state/action only
C4: Gripper	Franka parallel gripper	3D-printed hard gripper	3D-printed soft gripper
C5: Policy	ACT in LeRobot	π -style VLA policy	Diffusion policy

The morphological chart produces five leading module concepts, denoted C1–C5. C1 is a dual-FR3 platform; C2 is GELLO-based teleoperation; C3 is a synchronized three-camera, robot-state, and action pipeline; C4 is a 3D-printed soft gripper; and C5 is an ACT policy integrated through LeRobot. Together they form one integrated system concept, rather than five competing end-to-end robot systems. Their alternatives are retained because each exposes a real engineering trade-off: platform precision against integration cost, operator intuitiveness against mapping complexity, observation coverage against synchronization burden, compliance against geometric repeatability, and policy capability against data and deployment maturity. Detailed descriptions of the unselected alternatives are provided in the appendix.

3.2 Concept selection

3.2.1 Selection method and QFD-derived criteria

Selection was conducted in two stages. First, an absolute screen removed a candidate if it could not be made safe in the laboratory, could not interface with the selected robot and data path, or could not be obtained and integrated within the project schedule. Second, the surviving alternatives were scored on a five-point ordinal scale, where 1 denotes a weak fit and 5 denotes a strong fit. Ratings were assigned from the QFD relationship to ES1–ES6, laboratory availability and interface knowledge, the existing collection workflow, and the cited primary or official sources; no rating is presented as an unreported benchmark result. Consistent with the functional decomposition, the “top five concepts” are the five selected module concepts C1–C5. Their combination forms the single system concept carried into the final design. For candidate a in module m , the decision score is

$$S_{m,a} = \sum_{k=1}^4 w_k r_{m,a,k}, \quad (3-1)$$

where $r_{m,a,k}$ is the rating for criterion k and w_k is its QFD-derived weight. Scores are used only to compare alternatives within the same module; they are recorded design judgments, not measured task-performance results.

Table 3–2 QFD-derived criteria for module selection

No.	Criterion	Weight	QFD Link
SC1	Contribution to task performance and precision	0.30	ES1, ES2, ES3, ES4
SC2	Data, control, and software integration compatibility	0.25	ES1, ES2, ES3
SC3	Safety, robustness, and recovery support	0.20	ES5, ES6; CR3–CR4
SC4	Availability, manufacturability, and schedule feasibility	0.25	CR6–CR7 and project plan

3.2.2 Scoring of the five selected module concepts

Table 3–3 provides the detailed selection record for the five leading module concepts. The same criterion definitions and weights are applied in every module so that the scoring

logic remains traceable to the QFD; however, the totals are interpreted only within a module because a robot platform, a gripper, and a policy make different contributions to the integrated system.

Table 3–3 Module-level concept selection matrix (1 = weak fit; 5 = strong fit)

Module	Candidate	SC1	SC2	SC3	SC4	Weighted score
C1 Platform	Dual Franka FR3	5	5	4	3	4.30
	Dual Flexiv	4	3	4	2	3.30
	ALOHA-style bimanual	3	4	3	5	3.85
C2 Teleoperator	GELLO joint-master	4	5	4	5	4.50
	PICO 4 Ultra XR	3	3	3	4	3.25
	UMI hand-held	3	3	3	4	3.25
C3 Data collection	Two wrist + top camera + state/action	5	5	4	3	4.35
	One wrist + top camera + state/action	4	4	4	5	4.20
	Robot state/action only	2	3	3	5	3.25
C4 Gripper	3D-printed soft gripper	4	4	4	5	4.20
	Franka parallel gripper	3	5	4	3	3.70
	3D-printed hard gripper	4	5	3	4	4.05
C5 Policy	ACT in LeRobot	4	5	4	5	4.50
	Diffusion Policy	4	3	4	4	3.75
	π -style VLA policy (future route)	5	2	3	1	2.75

3.2.3 Comparative evaluation and selected concept

C1: Dual Franka FR3 platform. The dual Franka FR3 platform was selected as the mechanical foundation of the final system because it provides repeatable 7-DoF control, a direct connection to the available laboratory control ecosystem, and a credible path toward coordinated bimanual manipulation. Its principal disadvantage is the substantial integration effort required for workspace coordination, collision avoidance, commissioning, and bilateral control. Nevertheless, these costs are justified by the project objective: lower-cost alternatives

may reduce initial hardware burden, but they would introduce a different control ecosystem or provide a weaker fit to the required assembly precision.

C2: GELLO teleoperation. GELLO was selected as the planned demonstration-collection interface because it offers a direct and intuitive connection between operator motion and the robot-state/action records required by the learning pipeline^[18]. Despite the need for reliable leader-follower mapping and calibration, GELLO remains preferable to immersive or hand-held alternatives for the present system because it provides the most direct route from tele-operated demonstrations to the joint-space data representation used by the FR3 and LeRobot workflow.

C3: Three-camera data pipeline. The selected sensing configuration uses one Intel RealSense D435 overhead camera and two Intel RealSense D405 wrist-mounted cameras, together with synchronized robot state, gripper state, and action records. The D435 provides a stable global view of the shared workspace, while the two D405 cameras preserve local visual information near the end-effectors and contact regions. This complementary arrangement reduces visual blind spots and directly supports the component-detection and pose-estimation requirements in ES3–ES4, at the cost of increased synchronization and data-management effort.

C4: 3D-printed soft gripper. The selected end-effector is a 3D-printed soft gripper derived from the open-source gripper design in the referenced UMI material; the project adopts only the gripper-finger component rather than reproducing the complete UMI hardware system^[19]. Compliant contact and rapid geometry iteration are valuable for handling assembly components with small geometric variation, although deformation may reduce pose repeatability compared with a rigid locating gripper.

C5: ACT policy in LeRobot. ACT was selected as the current policy because it is integrated with the LeRobot workflow and supports task-specific training and checkpoint deployment from synchronized visual observations and robot-state inputs^[12,20]. Action chunking provides a practical temporal-control mechanism for the present imitation-learning pipeline. Deployment remains task-specific and operator-selected: the policy does not yet infer tasks from natural-language input or demonstrate unified multi-task control. This limitation is accepted because ACT provides a deployable and traceable baseline for the current project stage, whereas Diffusion Policy remains an earlier baseline and π -style VLA models remain

a future extension.

The selected architecture is the combination of the highest-scoring surviving candidate in each module. C1 and C4 establish the physical capability for controlled contact; C2 and C3 make demonstrations and visual evidence available to the learner; and C5 provides the deployable policy path. This combined decision covers the full ES1–ES6 chain while retaining a feasible implementation route for subsequent final design, manufacturing, and validation.

Chapter 4 Final design and implementation

4.1 Selected concept description

4.1.1 Design evolution from the DR3 baseline

The final implementation retains the modular physical architecture selected in Design Review 3 (DR3), but the learning system and demonstrated scope have changed substantially. DR3 proposed a dual-Franka Research 3 (FR3) workcell, GELLO demonstration collection, two wrist cameras and one overhead camera, custom compliant fingers, a LeRobot dataset, and an imitation-learning policy. At that stage, Action Chunking with Transformers (ACT) and task-specific checkpoints were the deployable baseline. Language-conditioned bimanual control was a future objective.

The completed system replaces the task-specific ACT controller with the LeRobot implementation of $\pi_{0.5}$. It uses two FR3 arms, two grippers, three RGB views, a 16-dimensional bilateral state and action interface, and one jointly trained multi-task checkpoint. The three tasks are selected by natural-language instructions rather than by manually loading three independent policy files. ACT therefore remains part of the design history and concept-selection evidence, but it is not presented as the final controller.

The mechanical elements selected in DR3 remain directly relevant. The two FR3 arms provide a common kinematic and controller interface, and the two GELLO leaders provide matched bilateral demonstration inputs. The wrist and overhead cameras provide complementary local and global views. The 3D-printed compliant fingers provide a task-compatible contact interface. The later policy update changes how these modules are connected and trained, but does not remove the need for their original engineering analysis.

4.1.2 Overall system architecture

The final system is organized as four connected layers:

1. the physical workcell, consisting of two FR3 arms, two custom grippers, task fixtures, and three cameras;
2. the demonstration layer, consisting of the two GELLO leader devices, the ROS 2 bridge, and the episode recorder;

3. the learning layer, consisting of LeRobot Dataset v3.0, the $\pi_{0.5}$ training configuration, and task-balanced validation; and
4. the deployment layer, consisting of language-conditioned policy inference, action-chunk handling, the bilateral robot adapter, and deterministic safety checks.

During demonstration collection, the operator moves the two GELLO leaders. The follower-side system commands the two FR3 arms. It records three RGB images, the measured bilateral state, the bilateral arm targets, the gripper commands, the episode index, and the task instruction. Accepted episodes are stored in LeRobot Dataset v3.0 format. During training, the three task datasets are sampled with equal task probability and used to update one $\pi_{0.5}$ model. During deployment, one of the three language instructions and the current three-view observation are supplied to the same checkpoint. The policy returns an action chunk, and the robot-side adapter sends the admissible absolute targets to the two FR3 controllers.

Figure 4–1 is retained from DR3 because it records the selected modular architecture and the intended connection between demonstration collection, learning, and closed-loop execution. Two parts of the diagram are historical: the ACT block has been replaced by $\pi_{0.5}$, and the task-specific checkpoint block has been replaced by one unified checkpoint. The three-camera workcell, GELLO collection concept, LeRobot dataset, action-chunk interface, dual-FR3 controller, and deterministic failure-handling boundary remain consistent with the final system.

4.1.3 Selected module concepts

4.1.3.1 Dual-FR3 bimanual platform

Two FR3 arms are used as the follower robots. The use of the same robot model on both sides reduces interface asymmetry: each side contributes seven joint positions, receives seven absolute joint targets, and uses the same low-level controller family. The shared kinematic structure also makes bilateral indexing, joint-limit validation, and data normalization easier to audit. Bimanual operation is required because each task includes a preparation or stabilization action followed by a more precise manipulation action.

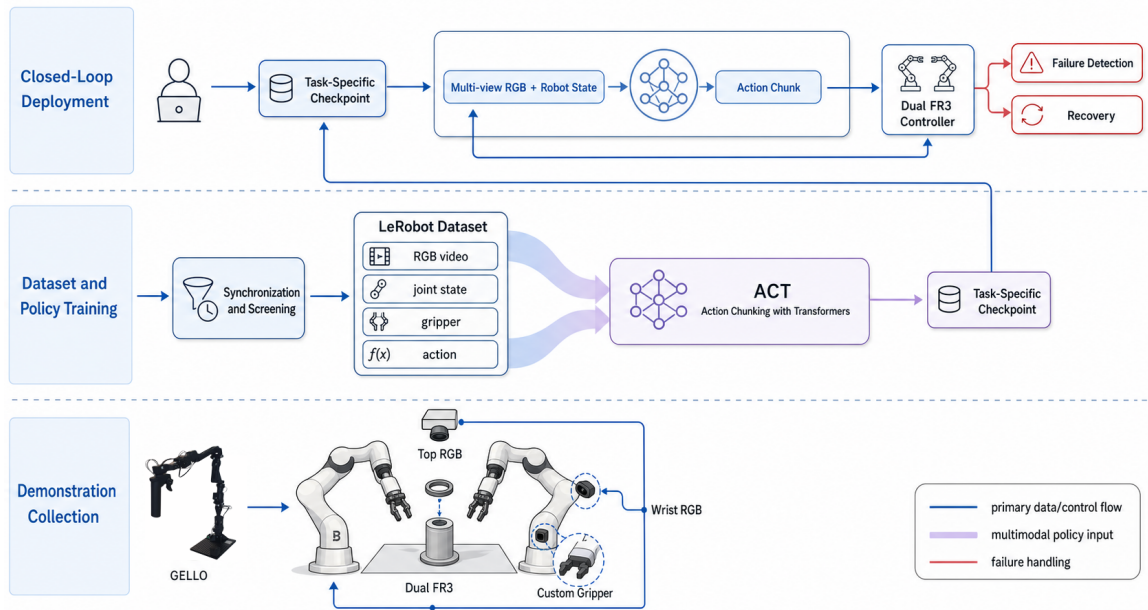


Illustration 4–1 System architecture retained from DR3 as the design baseline. In the final implementation, the ACT and task-specific-checkpoint blocks are superseded by one language-conditioned LeRobot $\pi_{0.5}$ checkpoint; the physical workcell and data-flow modules remain applicable.

4.1.3.2 GELLO demonstration interface

The demonstration interface uses one GELLO leader for each follower arm. GELLO provides a physical leader whose joint arrangement is mapped to the follower robot, giving the operator direct control of coordinated arm motion^[18]. The selected project configuration provides seven arm channels and one gripper channel per leader. Its low-cost modular construction also allows damaged or geometrically unsuitable leader components to be reprinted without changing the follower robot.

4.1.3.3 Three-camera visual observation

The final observation contains one fixed overhead view and two wrist views. The overhead Intel RealSense D435i observes the shared workspace, fixture positions, and large object motion. The two Intel RealSense D405 cameras move with the left and right wrists and preserve local visual information near grasp, contact, insertion, extraction, and placement regions. This arrangement avoids relying on a single global view that may be occluded by

either arm, while avoiding a wrist-only configuration that loses global task context.

4.1.3.4 Custom compliant grippers

Each arm uses the custom gripper developed during DR3. An assembly contains two identical screw-mounted fingers printed in TPU 64D. Each finger has an overall bounding-box size of

$$123.36 \times 25.80 \times 40.00 \text{ mm.}$$

The tapered outline and repeated ribs permit passive deformation around the handled ring, the insulated plug body, the motherboard edge, and the memory module. The reinforced mounting end keeps the screw interface stiffer than the contact region. Compliance is used to increase geometric tolerance; it is not treated as a substitute for camera coverage, accurate demonstration, or robot-side contact limits.

4.1.3.5 LeRobot $\pi_{0.5}$ policy

The final policy is the LeRobot implementation of $\pi_{0.5}$, adapted from the open-source OpenPI implementation^[20-23]. It receives three RGB images, the current bilateral state, and a language instruction. It predicts a chunk of bilateral actions in the same absolute joint-and-gripper representation used by the recorder and executor. The same model parameters are used for all three tasks.

4.1.4 Operational workflow

4.1.4.1 Workcell initialization

Before an episode or an autonomous trial begins, the two arms are returned to the documented initial joint region, the grippers are inspected, and the task objects are reset. Camera mounts, image identities, network connections, controller status, workspace boundaries, and emergency-stop access are checked. The socket fixture is kept electrically unpowered. A run is not started if a camera is missing, a robot state is stale, or the scene does not match the selected instruction.

4.1.4.2 Demonstration collection

The operator performs the selected task using the two GELLO leaders. The ROS 2 bridge assembles packets containing the three images, the measured robot and gripper state, and the

commanded arm and gripper values. The recorder samples these packets at 5 Hz and stores a complete episode. The operator then reviews task completion and resets the workcell before the next episode. Episode boundaries are preserved so that complete trajectories, rather than individual frames, can be assigned to training or validation.

4.1.4.3 Dataset screening and training

The recorded episode is checked for the presence and consistent length of all three image streams, the 16-dimensional state, the 16-dimensional action, the task identity, and the episode metadata. Accepted episodes are added to one of the three task datasets. Training then samples the three tasks with equal probability, applies the defined image and numeric preprocessing, and fine-tunes one $\pi_{0.5}$ checkpoint for 100,000 optimizer steps.

4.1.4.4 Closed-loop deployment

At deployment, the operator selects the natural-language instruction that matches the prepared scene. The model receives the instruction, three current images, and the current bilateral state. It predicts 50 actions, of which the first 10 are consumed before a new observation is evaluated. The executor checks the returned targets and sends them to the two arms at the dataset-aligned control rate. A task terminates on completion, timeout, invalid observation, invalid action, protective stop, emergency stop, or manual intervention.

4.2 Engineering design analysis

4.2.1 Relevant engineering fundamentals

The implemented system combines serial-manipulator kinematics, shared-workspace bi-manual coordination, compliant contact, multi-view machine perception, imitation learning, packet-level data synchronization, and receding-horizon control. The learned policy produces high-level position targets. It does not replace the FR3 low-level servo controller, joint-limit enforcement, protective-stop behavior, or the external emergency-stop procedure.

For one seven-degree-of-freedom arm with joint vector

$$\mathbf{q}_t = [q_{1,t}, \dots, q_{7,t}]^T,$$

the end-effector pose is

$${}^B\mathbf{T}_{E,t} = f(\mathbf{q}_t), \quad (4-1)$$

where B is the robot-base frame, E is the end-effector frame, and $f(\cdot)$ is the forward-kinematics map. In the dual-arm system, the corresponding transforms are

$${}^{B_L}\mathbf{T}_{E_L,t} = f_L(\mathbf{q}_t^L), \quad {}^{B_R}\mathbf{T}_{E_R,t} = f_R(\mathbf{q}_t^R). \quad (4-2)$$

If both base poses are expressed in a common workcell frame W , the relative end-effector transform can be written as

$${}^{E_L}\mathbf{T}_{E_R,t} = ({}^W\mathbf{T}_{B_L} {}^{B_L}\mathbf{T}_{E_L,t})^{-1} {}^W\mathbf{T}_{B_R} {}^{B_R}\mathbf{T}_{E_R,t}. \quad (4-3)$$

Equation 4–3 explains why the two base installations and task fixtures must remain fixed: the learned policy observes images and joint states rather than an explicitly recalibrated relative pose at every episode.

4.2.2 Design assumptions and constraints

The system operates in a structured indoor workcell. Lighting, camera mounts, robot bases, and task fixtures remain sufficiently repeatable for the learned visual policy. Each trial begins within the demonstrated workspace. The can, handled ring, socket box, plug, motherboard, memory module, and yellow tray are treated as task-specific laboratory objects rather than arbitrary unseen industrial components.

The current model uses RGB images and bilateral proprioception. Although the selected RealSense devices can provide depth, and the D435i includes inertial sensing, depth and inertial measurements are not included in the implemented $\pi_{0.5}$ feature contract. The chapter therefore uses manufacturer depth ranges and fields of view to justify camera placement, but it does not claim that the policy performs explicit metric depth reconstruction or six-degree-of-freedom pose estimation.

The recorder relies on software packets assembled by an upstream ROS 2 bridge. It does not use a hardware trigger or exposure-level synchronization. It also does not perform device-time interpolation or ROS approximate-time alignment. The valid engineering claim is packet-level synchronization, not exact simultaneity of the three camera exposures.

The design also assumes that deterministic safeguards remain outside the policy. The learned action is accepted only after dimensional, finite-value, joint, gripper, workspace, freshness, and communication checks. The socket task uses an electrically unpowered fixture.

The system is a supervised research workcell and is not represented as a certified production cell.

4.2.3 FR3 parameters and bimanual kinematic suitability

Each FR3 provides seven degrees of freedom, a nominal payload of 3 kg, a maximum reach of 855 mm, and link-side torque sensing in all seven axes^[24-26]. These values are suitable for the project because the manipulated objects are light, the task is performed on a tabletop, and the seventh joint gives redundancy for changing wrist orientation while avoiding fixtures and the other arm.

Table 4–1 FR3 parameters relevant to the selected bimanual workcell

Parameter	Value	Design relevance
Degrees of freedom	7 per arm	Provides redundant approach configurations and wrist orientation control.
Nominal payload	3 kg	Exceeds the mass required for the project objects and custom fingers.
Maximum reach	855 mm	Supports a shared central tabletop region when the two bases are arranged on opposite sides.
Joint sensing	Link-side torque sensing in all seven axes	Supports compliant low-level control, contact monitoring, and protective behavior.
Repeatability	Better than ± 0.1 mm in the current manufacturer specification	Robot repeatability is not the limiting claim for the learned end-to-end task; fixture, visual, demonstration, and contact errors also contribute.
Mounting interface	DIN ISO 9409-1-A50	Provides a standardized mechanical interface for the gripper assembly.
Installation	Upright	Matches the tabletop workcell and the workspace drawings used during concept selection.

The nominal joint ranges are A1 and A3: -166° to 166° ; A2: -105° to 105° ; A4: -176° to -7° ; A5: -165° to 165° ; A6: 25° to 265° ; and A7: -175° to 175° ^[24]. These are hardware bounds rather than learned-policy targets. The practical task workspace is kept smaller so that action chunks do not intentionally approach mechanical or controller limits.

Figure 4–2 records the arm geometry and principal link dimensions used during DR3. Figure 4–3 shows the manufacturer workspace envelope. The 855 mm maximum reach is not treated as an invitation to place objects at the boundary: motion near maximum extension reduces available orientation freedom and can bring the arm closer to singular configurations.

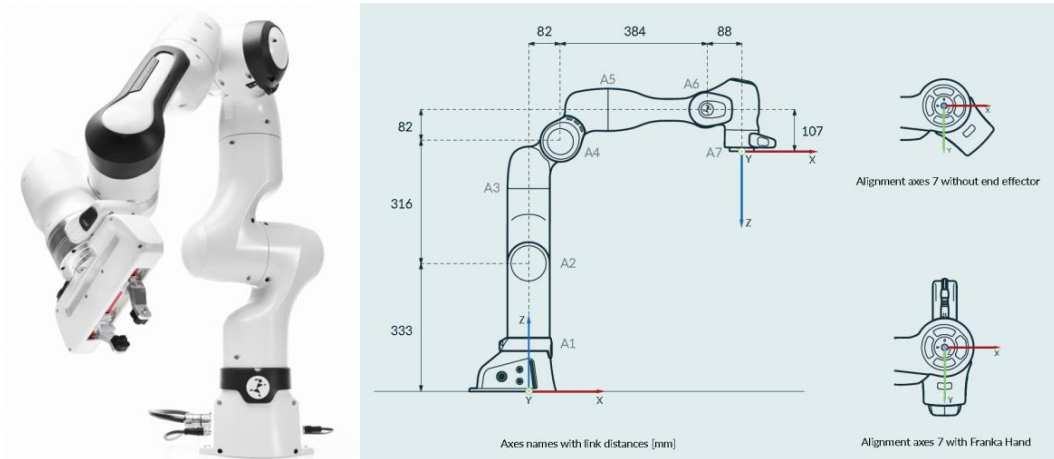


Illustration 4–2 FR3 arm and principal kinematic dimensions retained from the DR3 hardware analysis.

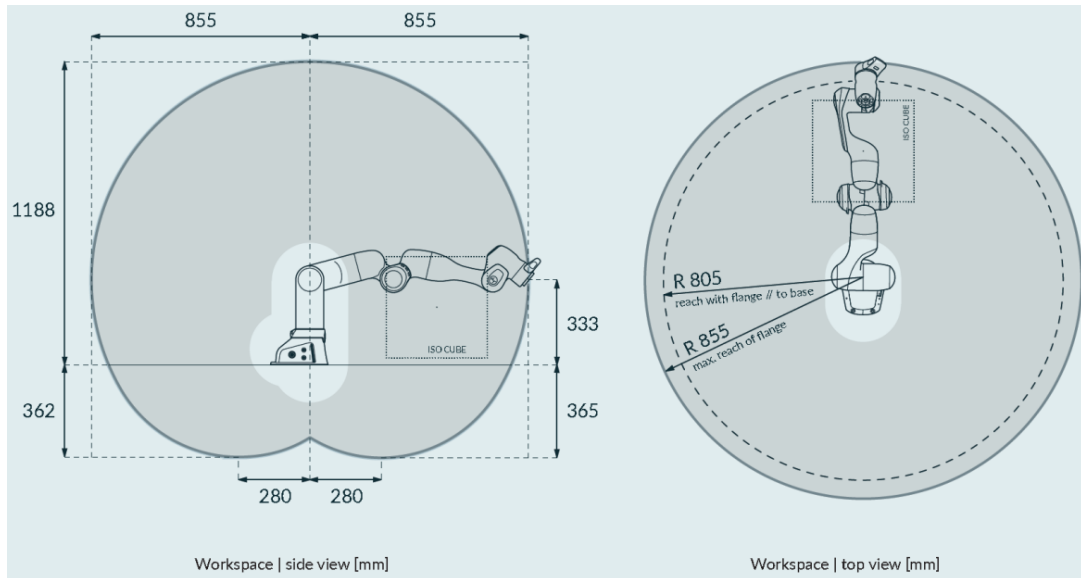


Illustration 4–3 Side and top views of the FR3 workspace envelope. The shared task region is selected inside the overlap of the two practical arm workspaces rather than at the 855 mm reach boundary.

ARM			
Degrees of freedom	7	Interfaces	• ethernet (TCP/IP) for visual intuitive programming with Desk
Payload	3 kg		• safety-rated input for external enabling device
Maximum reach	855 mm		• 2 configurable safety-rated inputs for emergency stop devices, safeguards or other protective devices (OSSD devices via external OSSD converter connectable)
Force/Torque sensing	link-side torque sensor in all 7 axes		• hardware prepared for: 2x DI & 2x DO (24V, isolated, EN 61131-2 type 3 characteristics, 100 Hz sampling rate)
Joint position limits	A1, A3: -166/166 deg A2: -105/105 deg A4: -176/-7 deg A5: -165/165 deg A6: 25/265 deg A7: -175/175 deg		• Control connector • connector for end effector
Mounting flange	DIN ISO 9409-1-A50	User Interfaces at the Arm's Pilot Grip	• integrated safety-rated guiding enabling switch
Installation position	upright		• guiding button
Weight	~ 17.8 kg		• guiding mode selector
Protection rating	IP40	User Interfaces at the Arm's Pilot Disc	• status light
Ambient temperature ²	+5 °C to +45 °C		• Pilot mode selector
Air humidity	20 ~ 80 % non-condensing		• arrow keys, teach, confirm, delete

Illustration 4–4 FR3 arm, joint limits, mounting, environmental, and controller-interface parameters documented during DR3.

4.2.4 Bimanual state and action representation

The actual state and action are both stored as 16-dimensional `float32` arrays. The state is

$$\mathbf{s}_t = [q_{1,t}^L, \dots, q_{7,t}^L, g_t^L, q_{1,t}^R, \dots, q_{7,t}^R, g_t^R]^\top \in \mathbb{R}^{16}, \quad (4-4)$$

where g_t^L and g_t^R are measured gripper opening widths in metres. The action uses the same ordering:

$$\mathbf{a}_t = [q_{1,t+1}^{L,target}, \dots, q_{7,t+1}^{L,target}, g_t^{L,target}, q_{1,t+1}^{R,target}, \dots, q_{7,t+1}^{R,target}, g_t^{R,target}]^\top. \quad (4-5)$$

Indices 0–6 correspond to the left arm, index 7 to the left gripper, indices 8–14 to the right arm, and index 15 to the right gripper. Arm actions are absolute next-step joint targets, not joint increments. Gripper actions are absolute `open_width_m` commands. The implemented setting is therefore `use_relative_actions=false` in both training and deployment.

The $\pi_{0.5}$ implementation pads state and action features to an internal maximum of 32 dimensions. Only entries 0–15 contain project data. Entries 16–31 are padding and are never interpreted as additional robot commands. The robot adapter explicitly slices and validates the first 16 entries before decomposing them into left-arm, left-gripper, right-arm, and right-gripper targets.

4.2.5 Shared workspace and role allocation

The two arms operate in a shared central region, so task role allocation is selected to reduce unnecessary crossing. In Task 1, the left arm pushes the can toward the centre while the right arm grasps and places the handled ring. In Task 2, the right arm pushes the socket box while the left arm grasps and inserts the plug. In Task 3, the left arm grips and pulls the motherboard while the right arm removes the memory module and places it in the yellow tray.

These assignments separate the large preparation motion from the subsequent precision motion. They also preserve the view of the precision-side wrist camera: the wrist that performs the ring placement, plug insertion, or RAM extraction approaches the relevant contact region directly. Role allocation is encoded by the demonstrations and the fixed action indexing; it is not dynamically reassigned by a separate planner.

The workcell dimensions and exact base-to-fixture transforms were not supplied as final measured values. Consequently, this report does not invent a dimensioned collision-free volume. The implemented engineering rule is that fixtures are placed in an experimentally verified overlap region, nominal trajectories stay away from the manufacturer reach boundary, and the robot-side workspace limits are tighter than the theoretical workspace shown in Figure 4–3. A production release would require the final base poses, fixture coordinates, swept-volume analysis, and inter-arm collision geometry to be documented numerically.

4.2.6 Contact, clearance, and insertion tolerance analysis

4.2.6.1 Ring-over-can placement

The ring task contains three mechanically distinct phases: a planar push of the can, a grasp of the ring handle, and a vertical placement of the ring over the can. Successful placement requires the ring inner opening to remain clear of the can outer profile throughout descent.

If D_r is the effective ring inner diameter and D_c is the effective can outer diameter, the ideal radial clearance is

$$c_r = \frac{D_r - D_c}{2}. \quad (4-6)$$

The allowable lateral placement error must be smaller than c_r , with an additional margin for ring tilt, finger deformation, and uncertainty in the apparent object boundary. The actual D_r and D_c values were not provided, so Equation 4–6 defines the required measurement rather than claiming a numerical tolerance.

The handle allows the right gripper to avoid pinching the ring circumference, but it introduces an offset between the grasp point and the ring centre. This offset creates a moment that can tilt the ring during transport. A near-vertical final approach and release after the ring is supported by the target reduce the sensitivity to that moment.

4.2.6.2 Plug insertion

Plug insertion is a constrained contact task. Lateral and angular misalignment can produce edge contact, jamming, incomplete seating, socket-box displacement, or excessive force. For an insertion depth L_i and effective lateral clearance c_i , a first-order small-clearance angular bound is

$$\theta_{\max} \approx \tan^{-1} \left(\frac{c_i}{L_i} \right). \quad (4-7)$$

The relation shows that angular tolerance decreases as insertion depth increases. The socket box is pushed into a repeatable central region before insertion so that the left arm can approach with sufficient remaining wrist orientation range. The right-arm push must end before the left-arm precision approach enters the same local region.

The DR3 socket mock-up was electrically unpowered and had a recorded overall size of $80 \times 80 \times 30$ mm. The current task continues to use an unpowered socket fixture. The exact dimensions of the final socket box, pin clearance, insertion depth, and fixture compliance should be remeasured if the physical object differs from the DR3 mock-up. Until those values are recorded, success is defined functionally: the correct plug enters the correct socket, remains seated without manual assistance, and produces no visible damage.

4.2.6.3 Memory-module extraction

The motherboard task requires fixture stabilization, connector release, module extraction, and safe placement. The left gripper pulls the motherboard closer and holds or stabilizes it.

The right gripper must approach the memory module without contacting nearby components, apply a controlled extraction motion consistent with the connector geometry, and transport the released module to the yellow tray.

The mechanical tolerance is governed by the free space around the memory module, the width and compliance of the custom fingers, and the connector's extraction direction. Excess lateral motion can bend the module or load adjacent board components. Because connector retention force, neighbouring-component clearance, and module dimensions were not supplied, the report does not infer extraction force or a numerical clearance. These quantities remain required measurements for a complete contact-mechanics verification.

4.2.7 Custom gripper geometry and mechanical analysis

Each gripper assembly uses two identical TPU 64D fingers, and the dual-arm workcell therefore uses four fingers. For one finger,

$$L = 123.36 \text{ mm}, \quad T = 25.80 \text{ mm}, \quad H = 40.00 \text{ mm}.$$

The principal geometric ratios are

$$\frac{L}{T} = \frac{123.36}{25.80} \approx 4.78, \quad \frac{L}{H} = \frac{123.36}{40.00} \approx 3.08. \quad (4-8)$$

These ratios describe an elongated finger rather than a short rigid jaw. The repeated ribs reduce local bending stiffness along the contact region, the tapered outline improves access, and the thicker mounting end distributes screw load.

The same compliance has a trade-off. It accommodates object-shape and pose variation and can reduce concentrated contact pressure, but deformation can reduce grasp repeatability and make the effective fingertip pose load dependent. The task design therefore uses compliance for tolerance while retaining fixed camera identities, repeatable starts, and arm-side position control. No finite-element stress, fatigue-life, or calibrated force-deflection result is claimed because print orientation, infill, wall count, actual TPU properties, contact loads, and cycle count were not supplied.

4.2.8 Perception working distance, field of view, and camera selection

The overhead camera is an Intel RealSense D435i. Its specified ideal depth range is approximately 0.3–3 m and its depth field of view is approximately $87^\circ \times 58^\circ$ ^[27]. This makes

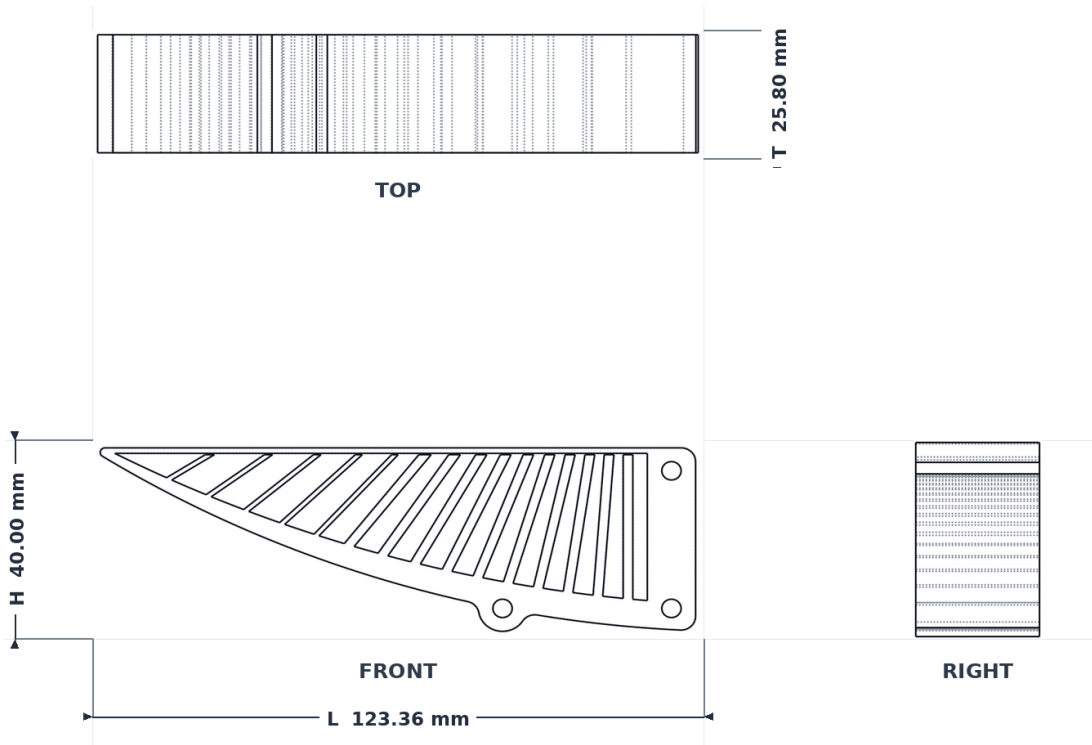


Illustration 4–5 Orthographic views of one TPU 64D compliant gripper finger, including the overall $123.36 \times 25.80 \times 40.00$ mm bounding-box dimensions.

it suitable for observing the complete tabletop from a fixed elevated position. The overhead view supplies global information when either wrist view is occluded or too close to show the relationship between the object, fixture, and destination.

Each wrist camera is an Intel RealSense D405. Its specified ideal depth range is approximately 7–50 cm, with an approximately $87^\circ \times 58^\circ$ depth field of view^[28]. This range is better matched to a camera mounted near the end effector. The wrist views preserve local detail during ring descent, plug alignment, and memory-module extraction.

The hardware can operate at higher image and depth rates than the learning pipeline, but the project records RGB at 424×240 and 5 Hz. The selected policy does not consume depth or D435i inertial data. Camera range and field of view therefore justify coverage and mounting, while the learned input remains three RGB images.

Table 4–2 Camera configuration and selection basis

Camera	Quantity	Nominal range	ideal	Depth FOV	Project function
RealSense D435i	1	0.3–3 m		$87^\circ \times 58^\circ$	Fixed overhead context for the shared workcell, fixtures, destinations, and large object displacement.
RealSense D405	2	7–50 cm		$87^\circ \times 58^\circ$	Left- and right-wrist close-range views for grasp, contact, insertion, extraction, and placement.

4.2.9 Image format and preprocessing

All three raw streams have width 424 and height 240. The LeRobot dataset stores each decoded RGB image in channel-first form:

$$[C, H, W] = [3, 240, 424].$$

Before an image is passed to $\pi_{0.5}$, it is converted to a $3 \times 224 \times 224$ tensor. The conversion preserves aspect ratio. The 424×240 image is scaled to approximately 224×126 , and the remaining square area is filled with approximately 49 black pixels above and below the resized image. The image is not stretched to 224×224 .

Pixel values are represented in $[0, 1]$ after decoding and are normalized to $[-1, 1]$ before the model. This process is applied independently but identically to the overhead, left-wrist, and right-wrist images. Camera names and ordering are part of the feature contract. A swapped wrist-camera mapping can produce a dimensionally valid but semantically incorrect policy input, so identity is checked before recording and deployment.

4.2.10 GELLO hardware and leader–follower mapping

One GELLO–Franka leader provides seven arm degrees of freedom and one gripper-related channel. The DR3 build specifies seven DYNAMIXEL XL330-M288-T actuators for

the arm joints and one XL330-M077-T actuator for the gripper-related joint, together with a ZNJ integrated control board, a 5 V 5 A switched-mode power supply, an FPX330-H101 structural kit, and 3D-printed leader and gripper-interface components.

The dual-arm system uses two such leader assemblies. At the component level this corresponds to fourteen XL330-M288-T units, two XL330-M077-T units, two controller boards, and two power supplies. The XL330 series operates from 3.7–6 V, with 5 V recommended; one actuator is approximately $20 \times 34 \times 26$ mm and approximately 18 g^[29]. The M288 and M077 variants trade speed and torque through different gear ratios. For this project, the leader devices provide position input rather than follower-scale actuation.



Illustration 4–6 GELLO–Franka leader device used as the basis of bilateral demonstration collection. The final system uses one leader assembly for each FR3 follower arm.

Leader motion must be mapped to follower joint order, sign convention, zero offset, and permitted range. An incorrect offset can generate a valid numeric command that is unsafe for the follower. The pre-collection procedure therefore verifies the leader zero, moves each channel through a small range, confirms the corresponding follower direction, and checks the gripper-width mapping before a task episode is recorded.

4.2.11 Data-collection pipeline

The implemented data path is

two GELLO leaders → two FR3 follower controllers → ROS 2 bridge
 → assembled packets
 → LeRobot recorder → Parquet records and three RGB videos.

The follower-side system provides measured joint positions and current gripper widths. The command side provides absolute arm targets and absolute gripper-width commands. The recorder samples the assembled packet stream at 5 Hz, assigns frames to the active episode, and preserves the task identity. A complete episode contains a continuous visual and proprioceptive record from the initial condition to either task completion or termination.

Collection and task review are intentionally separate operations. The recorder verifies schema and continuity, while the operator verifies whether the physical task was demonstrated correctly. A numerically complete episode can still be unsuitable for training if the operator intervened incorrectly, the object left the intended workspace, or the terminal state does not match the language instruction.

4.2.12 Packet synchronization and action-label alignment

The stored cadence is 5 Hz, corresponding to approximately 0.2 s per step. The recorder pairs adjacent assembled packets. For the saved sample at index t ,

$$\mathbf{o}_t^{\text{visual}} = \text{three images from packet } t, \quad (4-9)$$

$$\mathbf{s}_t = \text{bilateral measured state from packet } t, \quad (4-10)$$

$$\mathbf{a}_t^{\text{arm}} = \mathbf{q}_{t+1}^{\text{target}} \text{ from packet } t + 1, \quad (4-11)$$

$$\mathbf{a}_t^{\text{grripper}} = \mathbf{g}_t^{\text{target}} \text{ from packet } t. \quad (4-12)$$

Images and measured state therefore come from the same current packet. The arm label is shifted forward by one packet so that the current observation is paired with the next absolute joint target. The gripper label follows the existing same-packet convention.

This procedure guarantees consistency of the saved software packet and label rule. It does not prove that all three camera exposures occurred at the same physical instant. The

recorder does not interpolate device timestamps, perform ROS approximate-time alignment, or use a hardware trigger. It trusts the upstream bridge to provide an assembled packet. This limitation must be retained in the report because a nominal 5 Hz LeRobot timestamp alone cannot be used to reconstruct true camera skew or end-to-end latency.

4.2.13 Dataset construction and episode split

The three task datasets contain 80 episodes each. All three use random seed 1000 and the same validation episode indices:

$$\{17, 35, 58, 65, 66, 72, 74, 75\}.$$

The remaining 72 episodes per task are used for training. Splitting by episode prevents neighbouring frames from the same physical trajectory from appearing in both the training and validation sets.

Table 4–3 Episode-level dataset split for the three bimanual tasks

Task dataset	Episodes	Frames	Training split	Validation split
dual_ring	80	46,326	72 episodes / 41,410 frames	8 episodes / 4,916 frames
dual_socket	80	38,511	72 episodes / 34,655 frames	8 episodes / 3,856 frames
dual_ram	80	34,813	72 episodes / 31,675 frames	8 episodes / 3,138 frames
Total	240	119,650	216 episodes / 107,740 frames	24 episodes / 11,910 frames

The tasks are sampled with equal probability during training. Task selection occurs before sample selection, so `dual_ring` does not receive a larger task-level probability merely because it contains more frames. This choice balances the three instructions and prevents the longest average trajectory from dominating the gradient updates.

4.2.14 $\pi_{0.5}$ base model and software configuration

The final controller uses the official `pi05_base` model. The upstream checkpoint is identified by `gs://openpi-assets/checkpoints/pi05_base/params`, and training reads the converted PyTorch parameters from `real-exp/.hf-cache/pi05_base_pytorch/model.safetensors`.

The base architecture combines a PaliGemma gemma_2b vision-language component with a gemma_300m action expert. The model weights use `bfloat16`.

The project uses LeRobot 0.5.2 with local project modifications. The base commit is

`dc06dba9d6220ab99c94e83ec7f0d45a35b44bed`.

The dataset format is LeRobot Dataset v3.0. Training is executed on physical GPU 1 of an NVIDIA RTX A6000 with 48 GB of memory.

4.2.15 Training procedure

The three tasks are trained jointly for 100,000 optimizer steps. Batch size is 4 and the optimizer is AdamW. The peak learning rate is 2.5×10^{-5} . A 1,000-step warm-up is followed by cosine decay to 2.5×10^{-6} . Task-balanced validation is run every 5,000 steps.

Table 4–4 Final $\pi_{0.5}$ training configuration

Parameter	Implemented value
Base model	Official pi05_base
Model structure	PaliGemma gemma_2b and action expert gemma_300m
Weight precision	<code>bfloat16</code>
LeRobot version	0.5.2, with local modifications
Dataset format	LeRobot Dataset v3.0
Training hardware	NVIDIA RTX A6000 48 GB, physical GPU 1
Batch size	4
Optimizer	AdamW
Training length	100,000 joint-training steps
Learning-rate schedule	1,000-step warm-up to 2.5×10^{-5} , followed by cosine decay to 2.5×10^{-6}
Task sampling	Equal probability for <code>dual_ring</code> , <code>dual_socket</code> , and <code>dual_ram</code>
Validation interval	Every 5,000 optimizer steps

The training configuration deliberately separates task balancing from frame count. The action and state normalizers are derived from the training data only. Validation episodes are not used to compute model updates. The same action representation and `use_relative_actions=false` setting are retained when the selected checkpoint is loaded for deployment.

4.2.16 Validation and checkpoint selection

For each task, validation first computes the mean normalized flow-matching loss over at most 64 batches. With batch size 4, this corresponds to at most 256 validation samples per task at one validation point. The checkpoint-selection quantity is the unweighted task macro-average

$$\mathcal{L}_{\text{macro}} = \frac{\mathcal{L}_{\text{ring}} + \mathcal{L}_{\text{socket}} + \mathcal{L}_{\text{ram}}}{3}. \quad (4-13)$$

The best checkpoint is the validation point with the lowest $\mathcal{L}_{\text{macro}}$.

This metric is intentionally not weighted by the number of validation frames. It gives each task equal influence, just as the training sampler gives each task equal selection probability. The loss supports checkpoint comparison, but it is not a physical success rate, insertion-depth measurement, contact-force measurement, or calibrated pose error. Real-robot performance is evaluated separately in Chapter 5.

4.2.17 Action chunking and temporal interface

Collection and execution use 5 Hz, so one action step represents 0.2 s. The model predicts a 50-step chunk:

$$T_{\text{predicted}} = \frac{50}{5} = 10 \text{ s}. \quad (4-14)$$

Deployment uses `n_action_steps=10`, so the executor consumes approximately

$$T_{\text{executed}} = \frac{10}{5} = 2 \text{ s} \quad (4-15)$$

before requesting a new policy result from updated observations.

This is a receding-horizon interface. The 50-step prediction gives the action expert a longer temporal target, while executing only 10 steps limits the amount of open-loop motion. The exact policy-query interval may also include capture, preprocessing, inference, transfer, and queue overhead; therefore, the nominal two-second value describes the consumed action segment rather than a measured end-to-end latency.

4.2.18 Deployment pipeline

Deployment begins by loading the selected unified checkpoint and its associated normalization statistics. The system verifies the three camera feature names, checks the 16-dimensional state, and confirms the language instruction. The current observation is prepro-

cessed and passed to the policy server. The returned action chunk is denormalized, sliced to the 16 valid robot dimensions, checked for finite values and limits, and divided into left-arm, left-gripper, right-arm, and right-gripper targets.

The robot adapter sends absolute joint targets and absolute gripper widths. It does not reinterpret them as deltas. After the configured ten-step segment, new images and state are acquired and a new action chunk is requested. This loop continues until the terminal task condition or a stop condition is reached.

4.2.19 Control, computation, and safety boundary

The policy is not the safety controller. Before motion is enabled, a dry run verifies camera freshness and ordering, state/action dimensions, normalization statistics, checkpoint identity, language instruction, finite model outputs, network communication, joint and gripper limits, workspace bounds, and emergency-stop access.

During motion, the run stops on a missing image, stale state, malformed packet, non-finite output, excessive target, workspace violation, communication timeout, robot protective stop, emergency stop, or manual stop. Reset is performed only after policy execution has stopped and the shared workspace is clear. The power socket remains unpowered. These checks maintain a deterministic boundary around the probabilistic learned policy.

4.3 Detailed system design

4.3.1 Workcell layout

The two FR3 arms are mounted upright on opposite sides of a shared tabletop region. Task fixtures are placed inside the verified overlap of the two practical workspaces. The D435i is fixed above the shared region. One D405 is attached to each wrist. Each arm carries one custom gripper. The two GELLO leaders remain outside the follower workcell and are used by the operator during demonstration collection.

The lower row of Figure 4–1 records the selected physical arrangement: dual FR3 arms, a top view, wrist views, and custom grippers. The final setup extends the depicted single GELLO symbol to two leader assemblies. Exact base spacing, camera height, camera pitch, and fixture coordinates were not included in the supplied implementation record; these values

should be added to a future dimensioned workcell drawing rather than inferred from the conceptual image.

4.3.2 Dual-arm robotic platform

Both arms use the same 16-value bilateral ordering defined in Equations 4-4 and 4-5. The left and right controllers remain independent low-level systems, but they are driven from one policy output and one episode timeline. Each arm can perform a large scene-preparation motion or a precise manipulation motion, depending on the demonstrated task role.

The redundant seventh joint is important in the shared workcell because end-effector pose alone does not determine a unique arm posture. Demonstrations implicitly select postures that provide camera visibility, avoid fixtures, and reduce inter-arm interference. The low-level controller remains responsible for executing the admissible targets; the learned model does not directly output torque.

4.3.3 GELLO leader assemblies

Each GELLO leader contains the following hardware groups:

1. seven XL330-M288-T actuator modules for the arm joints;
2. one XL330-M077-T module for the gripper-related channel;
3. one ZNJ integrated control board;
4. one 5 V, 5 A switched-mode power supply;
5. one FPX330-H101 structural kit; and
6. the required 3D-printed leader links, brackets, base, and Franka-compatible gripper interface.

The control board reads the leader joint configuration and publishes targets to the follower-side interface. The operator does not bear follower-arm load through the leader; the leader is a measurement and command device. Mechanical looseness, servo zero error, cable interference, or incorrect joint signs can degrade demonstration quality, so the two leader assemblies are inspected and checked independently before bilateral collection.

4.3.4 End-effector and mechanical components

Each gripper uses two screw-mounted compliant fingers. The mounting screws must be tightened consistently, and the fingers must be installed with matching orientation. Before a run, the operator checks for tearing at rib roots, permanent bending, loose mounting holes, surface contamination, and insufficient opening range.

The task fixtures prevent uncontrolled object drift while retaining the manipulation required by the task. The can and socket box must still be pushable into the intended central region. The motherboard must be graspable and movable by the left gripper but sufficiently supported to allow the right gripper to extract the memory module. The tray must remain fixed so that successful placement is not confused with tray motion.

4.3.5 Vision and sensing system

The D435i mount is fixed relative to the workcell and is positioned to keep all source and destination regions in view. The two D405 mounts move with the wrists and are oriented so that the gripper contact region appears in the close-range image during approach. Cables are routed to avoid entering the gripper opening or restricting wrist motion.

The logical camera mapping is fixed:

$$\begin{aligned} \text{overhead} &\equiv \text{D435i}, \\ \text{left_wrist} &\equiv \text{left D405}, & \text{right_wrist} &\equiv \text{right D405}. \end{aligned}$$

The recorder and policy use these identities consistently. The three cameras provide RGB to the model; their depth and inertial capabilities are not part of the current learned observation.

4.3.6 Demonstration data-collection system

The collection system starts the two robot interfaces, both gripper interfaces, three camera publishers, the ROS 2 bridge, and the LeRobot recorder. The bridge assembles the latest required fields into a packet. The recorder begins a named task episode only after all required fields are available.

The operator executes the complete instruction and then ends the episode. The episode is retained only if the three image streams decode correctly, state and action arrays have the required width, and the demonstration matches the intended task. Manual workcell reset is

performed between episodes. The same task instruction is used during collection and later deployment so that language conditioning does not depend on an undocumented paraphrase.

4.3.7 Data processing pipeline

Processing has separate visual and numeric branches. The visual branch decodes the three RGB streams, converts them to channel-first tensors, preserves aspect ratio during resizing, pads to 224×224 , and normalizes to $[-1, 1]$. The numeric branch reads the 16-dimensional state and action, applies the training statistics, and pads the model-side representation to 32 dimensions.

Episode metadata determines the task and split. The eight validation episode indices are excluded from training for every task. During data loading, the task-balanced sampler chooses among the three task datasets with equal probability. The language instruction, three images, state, and supervised action target are then passed to the policy training step.

4.3.8 Policy-training system

The training system loads `pi05_base`, the converted PyTorch weights, and the LeRobot v3.0 datasets. It uses batch size 4 on the RTX A6000. Optimizer state, learning-rate schedule, model state, step index, and validation quantities are saved at the configured checkpoints.

Validation runs every 5,000 steps. Each task is evaluated separately before the unweighted macro-average is calculated. This preserves per-task traceability and prevents a low loss on the largest dataset from concealing a poor result on another task.

4.3.9 Robot control and deployment system

The deployment system loads one checkpoint for all three instructions. The selected instruction and current observation are sent to $\pi_{0.5}$. A 50-step action chunk is returned, and the executor uses the first ten steps. The target ordering, absolute-action interpretation, 5 Hz cadence, and camera mapping are identical to training.

The left and right arm targets are separated only after policy output validation. The gripper commands are expressed as absolute opening widths in metres. After each executed segment, the queue is not blindly extended with the rest of the old 50-step chunk; the policy is queried again using updated images and state.

4.3.10 Hardware–software interfaces

Table 4–5 Detailed hardware–software interfaces in the final system

Information	Source	Destination	Implemented contract
Left leader joints	Left GELLO	Left follower interface	Seven leader positions mapped to left FR3 joint order, signs, offsets, and permitted range.
Left leader gripper	Left GELLO	Left gripper interface	One leader channel mapped to absolute gripper opening width.
Right leader joints	Right GELLO	Right follower interface	Seven leader positions mapped to right FR3 joint order, signs, offsets, and permitted range.
Right leader gripper	Right GELLO	Right gripper interface	One leader channel mapped to absolute gripper opening width.
Left measured joints	Left FR3 controller	ROS 2 bridge / recorder / executor	Seven current joint positions in state indices 0–6.
Left measured gripper	Left gripper	ROS 2 bridge / recorder / executor	Current opening width in metres at state index 7.
Right measured joints	Right FR3 controller	ROS 2 bridge / recorder / executor	Seven current joint positions in state indices 8–14.
Right measured gripper	Right gripper	ROS 2 bridge / recorder / executor	Current opening width in metres at state index 15.
Overhead RGB	RealSense D435i	Bridge / recorder / policy	RGB 424×240 at 5 Hz; stored as $3 \times 240 \times 424$, then resized and padded to $3 \times 224 \times 224$.
Left-wrist RGB	Left RealSense D405	Bridge / recorder / policy	Same tensor contract under the fixed left-wrist feature name.
Right-wrist RGB	Right RealSense D405	Bridge / recorder / policy	Same tensor contract under the fixed right-wrist feature name.
Assembled packet	ROS 2 bridge	LeRobot recorder	Same-packet images and measured state; adjacent-packet arm action label.
Episode record	LeRobot recorder	Dataset v3.0	Three videos, state, absolute action, task instruction, episode and frame indexing.
Training sample	Dataset and sampler	$\pi_{0.5}$ trainer	Equal-probability task sampling; training episodes only.

Information	Source	Destination	Implemented contract
Language instruction	Task interface	$\pi_{0.5}$ policy	One of the three documented task instructions matching the prepared scene.
Policy observation	Preprocessor	$\pi_{0.5}$ policy	Three normalized $3 \times 224 \times 224$ images and a padded numeric state whose first 16 entries are valid.
Predicted action	$\pi_{0.5}$ policy	Robot adapter	50-step chunk; first 10 steps consumed per query.
Left arm command	Robot adapter	Left FR3 controller	Absolute joint targets from action indices 0–6.
Left gripper command	Robot adapter	Left gripper	Absolute opening width from action index 7.
Right arm command	Robot adapter	Right FR3 controller	Absolute joint targets from action indices 8–14.
Right gripper command	Robot adapter	Right gripper	Absolute opening width from action index 15.
Stop conditions	Cameras, bridge, adapter, robot controllers, operator	Both robot command paths	Stop on stale/missing data, invalid action, timeout, workspace or joint violation, protective stop, emergency stop, or manual stop.

4.3.11 Task definitions and bilateral role allocation

4.3.12 System operation

4.3.12.1 Pre-run checks

The operator confirms the checkpoint, instruction, object arrangement, camera mapping, gripper condition, network status, and robot status. Both arms are returned to the approved start region. The socket is verified to be unpowered. The shared workspace is cleared of unrelated objects, and the emergency stop remains accessible.

4.3.12.2 Collection run

The leader offsets and channel directions are checked. Recording begins only after all three cameras and all bilateral state/action fields are present. The operator executes the complete task through the two GELLO leaders. The episode is stopped, reviewed, accepted or

Table 4-6 Language instructions, bilateral roles, and terminal conditions

Task	Language instruction	Demonstrated arm roles	Terminal completion condition
T1	First, push the can to the middle, then pick up the black ring with the handle, put it over the can, and set it down.	Left arm pushes the can; right arm grasps the handled ring and places it.	The can is repositioned and the released ring remains stably over the can.
T2	Push the box with the power socket to the middle, then pick up the plug and insert it into the socket.	Right arm pushes the socket box; left arm grasps and inserts the plug.	The socket box is repositioned and the plug remains inserted without manual assistance.
T3	Pull the computer motherboard closer, then remove the memory module and place it in the yellow tray.	Left arm grips and pulls the motherboard; right arm removes and places the memory module.	The memory module is fully removed and remains inside the yellow tray.

rejected, and then the scene is reset.

4.3.12.3 Policy dry run

The policy is first queried without enabling robot motion. The returned shape, finite values, absolute-action interpretation, camera ordering, and initial target relative to the current robot state are inspected. A dry run that fails any check is not promoted to physical execution.

4.3.12.4 Autonomous run

The executor enables the approved command path, consumes ten actions from each predicted chunk, and re-observes the workcell. The operator does not manually assist a counted trial. The trial ends when the documented terminal condition is met or a stop condition occurs. Outcome, stop reason, checkpoint, instruction, and video record are retained for Chapter 5 evaluation.

4.3.13 Parts and components

Table 4–7 Parts and components for the final bimanual system

No.	Part or component	Specification	Qty.	Purpose
1	Franka Research 3 arm	7-DoF, 3 kg nominal payload, 855 mm maximum reach, joint torque sensing	2	Left and right follower manipulation
2	FR3 controller and robot interface	Manufacturer controller, network interface, low-level protection, FCI/ROS 2 integration	2 sets	State feedback and admissible joint-target execution
3	Custom gripper assembly	Project gripper with two screw-mounted compliant fingers	2	One end effector per arm
4	Compliant finger	TPU 64D, $123.36 \times 25.80 \times 40.00$ mm, tapered and ribbed	4	Object contact with passive geometric tolerance
5	Wrist camera	Intel RealSense D405, short-range RGB/depth device; RGB used by policy	2	Left- and right-wrist close-range observation
6	Overhead camera	Intel RealSense D435i; RGB used by policy	1	Global shared-workspace observation
7	GELLO leader assembly	7 arm channels plus 1 gripper channel	2	Bilateral human demonstration input
8	DYNAMIXEL XL330-M288-T	3.7–6 V actuator module; 5 V recommended	14	Seven arm joints per GELLO leader
9	DYNAMIXEL XL330-M077-T	3.7–6 V actuator module; 5 V recommended	2	Gripper-related channel on each GELLO leader
10	GELLO control board	ZNJ integrated control board	2	Read and communicate leader joint positions
11	GELLO power supply	5 V, 5 A switched-mode supply	2	Leader-device electrical supply
12	GELLO structural kit	FPX330-H101 with project 3D-printed links and brackets	2 sets	Leader mechanism and mounting
13	Camera mounts and cable routing	Fixed overhead mount and two wrist mounts	1 set	Preserve camera identity, pose, and cable clearance
14	Training workstation	NVIDIA RTX A6000, 48 GB; physical GPU 1 used	1	Joint $\pi_{0.5}$ fine-tuning and validation

No.	Part or component	Specification	Qty.	Purpose
15	Network and ROS 2 bridge host	Project-configured communication and packet-assembly system	1 set	Connect robots, grippers, cameras, recorder, and policy
16	LeRobot software environment	LeRobot 0.5.2 with local changes; Dataset v3.0	1 config.	Recording, preprocessing, training, and deployment
17	Can and handled ring	Task 1 objects	1 set	Repositioning, grasp, and ring placement
18	Socket box and plug	Electrically unpowered insertion task fixture and matching plug	1 set	Repositioning and constrained insertion
19	Motherboard, memory module, and tray	Task 3 objects; yellow destination tray	1 set	Stabilization, extraction, transport, and placement
20	Task fixtures and reset references	Printed or manually assembled positioning aids	3 sets	Repeatable task initialization and terminal-state inspection
21	Safety equipment	Physical emergency stop, manufacturer protective functions, supervised operation	1 set	Deterministic stop and operator protection
22	Data storage	Storage for 119,650 frames, three-view video, checkpoints, and logs	1 set	Dataset and experiment traceability

4.4 Manufacturing and implementation plan

4.4.1 Implementation scope

The project reuses laboratory robot arms, controllers, cameras, computing resources, and network infrastructure. Custom manufacturing is concentrated in the compliant fingers, GELLO structural parts, camera mounts, and task fixtures. Software implementation covers collection, packet assembly, LeRobot conversion, preprocessing, multi-task sampling, $\pi_{0.5}$ training, validation, inference, action adaptation, and run logging.

The full economic value of the workcell is not equal to the incremental cash expenditure. The FR3 arms, cameras, GPU, and safety equipment are in-kind laboratory resources. The DR3 budget recorded no additional out-of-pocket expense for the proof of concept. The expanded budget below preserves that statement while separating unpriced in-kind assets

from incremental fabrication cost.

4.4.2 Materials and manufacturing processes

Table 4–8 Materials, manufacturing processes, and critical considerations

No.	Part or resource	Material or source	Process or tool	Critical consideration
1	FR3 arms and controllers	Laboratory equipment	Upright installation and interface configuration	Base rigidity, shared workspace, controller limits, network isolation, emergency stop
2	Compliant fingers	TPU 64D	Fused-filament 3D printing, support removal, inspection	Print orientation, rib integrity, screw-hole quality, paired geometry, permanent deformation
3	Gripper attachment	Existing gripper and printed/adapted interfaces	Mechanical assembly with screws	Alignment, thread engagement, consistent tightening, opening clearance
4	GELLO structures	FPX330-H101 and printed parts	Additive manufacturing and modular assembly	Joint-axis alignment, low friction, cable strain relief, repeatable zero offsets
5	Camera mounts	Laboratory or printed brackets	Rigid mounting and cable routing	Working distance, FOV, gripper visibility, occlusion, cable interference
6	Task fixtures	Printed or manually assembled fixture parts	Additive manufacturing and bench assembly	Repeatable initial pose, pushability where required, no mains power, stable destination
7	Software environment	LeRobot, PyTorch, ROS 2, project modifications	Version-controlled configuration and scripted execution	Commit traceability, feature names, absolute-action setting, dataset/model compatibility
8	Dataset and checkpoints	Recorded demonstrations and training outputs	Validation, indexing, backup, and integrity checks	Episode-level split, three-view completeness, task balance, reproducible checkpoint selection

4.4.3 Gripper fabrication and inspection

The finger CAD is prepared for TPU 64D printing with the mounting face and rib structure preserved. The selected print orientation must avoid weak inter-layer loading at the mounting holes and rib roots. After printing, supports and stringing are removed without cutting the

compliant ribs. The overall envelope, paired symmetry, hole condition, and free deformation are inspected before assembly.

The two fingers on one gripper are installed with matching orientation and adequate clearance through the commanded opening range. A manual opening-and-closing check verifies that neither finger contacts the camera mount or wrist cable. The assembly is rejected if a rib is torn, the base is permanently warped, a screw cannot retain the finger, or the two contact surfaces are visibly mismatched.

4.4.4 GELLO assembly and calibration

Each leader is assembled from the FPX330-H101 structure, printed links and brackets, seven M288 actuators, one M077 actuator, controller board, and 5 V power supply. Joint axes are checked for free movement before wiring. Cables are routed so that the full demonstrated range does not pull on connectors or bias the leader posture.

After electrical assembly, servo identifiers, zero positions, directions, and range mappings are verified one channel at a time. The left and right leaders are calibrated independently because a mirrored workcell does not imply identical signs and offsets. The final mapping is stored with the collection configuration so that later episodes can be traced to the leader calibration used.

4.4.5 Camera and fixture implementation

The overhead mount is installed so that the complete source, interaction, and destination regions remain visible. Each D405 is mounted within its intended 7–50 cm close-range operating envelope during manipulation. The wrist view is checked at approach, contact, and retreat postures, not only at the robot start pose.

Task fixtures are placed in the verified shared workspace. The can, socket box, and motherboard must permit the instructed preparation motion, while their target regions remain visible. The yellow tray is fixed against accidental displacement. The socket fixture is electrically isolated and unpowered. Reference markings or stops are used where necessary to make scene reset repeatable.

4.4.6 Software implementation

The software stack is configured in independent stages:

1. robot, gripper, camera, and GELLO device interfaces;
2. ROS 2 state and command topics;
3. bridge packet assembly and freshness checks;
4. LeRobot Dataset v3.0 episode recording;
5. image preprocessing and 16-to-32-dimensional model padding;
6. episode-level split and equal-probability task sampling;
7. $\pi_{0.5}$ fine-tuning and macro-loss validation;
8. unified-checkpoint policy inference; and
9. bilateral absolute-action adaptation and safety checks.

The configuration records LeRobot version 0.5.2, the project commit, local modifications, base-checkpoint source, training steps, learning-rate schedule, camera feature names, `use_relative_actions=false`, action chunk size, and `n_action_steps`. These values are treated as part of the reproducible implementation rather than left as implicit code defaults.

4.4.7 System integration sequence

Integration proceeds from independent hardware checks to full learned control. First, each arm and gripper is checked without the policy. Second, all three cameras are checked under their final logical names. Third, each GELLO leader is mapped to its follower. Fourth, bilateral teleoperation is tested at low speed with an empty workspace. Fifth, complete episodes are recorded and decoded. Sixth, a training batch is inspected for camera order, numeric order, and action alignment. Seventh, the policy is run without opening the robot command path. Finally, supervised physical trials are enabled.

This sequence isolates failures. A camera-identity error should be found before training; a leader offset should be found before collecting a large dataset; a state/action index error should be found before robot execution; and a policy-output error should be found during the dry run.

Table 4–9 Expanded implementation budget and resource status

No.	Component or service	Provider	Qty.	Cash cost	Status and accounting note
1	Two FR3 arms, controllers, and safety equipment	Laboratory	2 sets	CNY 0 incremental	Existing in-kind assets; acquisition value not supplied
2	D435i and two D405 cameras with mounts	Laboratory	3	CNY 0 incremental	Existing in-kind assets; mount material not separately recorded
3	RTX A6000 training workstation and storage	Laboratory	1 set	CNY 0 incremental	Existing shared compute; energy and depreciation not allocated
4	Two GELLO leader assemblies	Laboratory / project resources	2	CNY 0 incremental	DR3 records availability; itemized acquisition value not supplied
5	TPU 64D compliant fingers	Student Innovation Center	4	CNY 0 incremental	Fabricated through available resources; material mass and print time not recorded
6	Task objects and fixtures	Laboratory / Student Innovation Center	3 sets	CNY 0 incremental	Available; final fixture-material value not supplied
7	Software stack	Open source and project code	1 config.	CNY 0 licence cost	LeRobot, ROS 2, PyTorch, and project modifications
Reported additional out-of-pocket cost				CNY 0	Excludes in-kind asset value, labour, energy, and depreciation

4.4.8 Budget

The CNY 0 value is therefore an incremental cash figure, not the total system value. A full economic budget would require the purchase or replacement value of two FR3 systems, two GELLO assemblies, three cameras, the GPU workstation, storage, safety hardware, print material, print time, labour, and energy. Those values were not included in the supplied DR3 or final implementation record and should be added only from invoices or the laboratory asset register.

4.4.9 Current implementation status and remaining documentation

The final implementation now includes dual-arm state and action handling, three RGB inputs, LeRobot Dataset v3.0 data, one jointly trained $\pi_{0.5}$ checkpoint, equal-probability three-task sampling, the defined episode-level validation split, macro flow-matching checkpoint selection, and a 5 Hz absolute-action deployment interface. The selected DR3 hardware modules and engineering drawings have been retained in this chapter.

Several quantities remain documentation gaps rather than model-design gaps. They include measured FR3 base poses, camera extrinsics and mounting distances, dimensioned fixture locations, ring and can clearances, and final socket insertion clearance and depth. They also include motherboard and memory-module clearances, measured gripper force–deflection behaviour, TPU print settings, print mass and time, full in-kind asset value, and measured end-to-end latency. These values should be added if the report is expected to make production-level tolerance, structural, timing, or economic claims.

Chapter 5 Validation results

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5.1 Validation overview

The validation procedure follows the DR3 engineering specifications in Chapter 2. The purpose of validation is to check whether the integrated system satisfies the intended proof-of-concept behavior under bounded laboratory conditions. The evaluated workflow includes scene setup, synchronized sensing, state acquisition, policy inference, robot and gripper command execution, and completion assessment.

The bimanual manipulation task has been completed on the integrated dual-arm setup. This result demonstrates that the selected hardware platform, teleoperation-derived data path, ACT-based policy module, and robot-side execution loop can be combined into a functioning bimanual manipulation system. The result should be interpreted within the defined project scope: the workspace, initial conditions, fixtures, and execution assumptions are controlled, and the system is not claimed to provide unrestricted open-world autonomy.

5.2 Key data

5.3 Validation summary

The completed validation confirms the main system-level objective of the project: a dual-arm FR3 manipulation platform can execute the scoped bimanual manipulation workflow using the selected data and policy architecture. The most important validation outcome is the integration of the complete chain from demonstration collection and dataset preparation to learned policy execution on the real robot. Detailed task descriptions, trial counts, and quantitative measurements are reserved for the blank key-data section.

Chapter 6 Discussion and conclusions

6.1 Discussion

This project developed a bimanual industrial manipulation system for representative 3C assembly operations. The final system integrates a dual-Franka Research 3 platform, GELLO-based demonstration collection, synchronized visual and proprioceptive data recording, task-compatible end effectors and fixtures, and an ACT-based policy deployment workflow. The design process remains traceable from customer requirements to engineering specifications, concept generation, concept selection, implementation, and validation.

A key outcome is that the bimanual manipulation task has been completed on the integrated real-robot setup. This confirms that the selected mechanical, sensing, data, and policy modules can operate together as a system rather than only as isolated components. The completed bimanual result also supports the original motivation of moving beyond single-arm demonstrations toward coordinated manipulation in a shared workcell.

The main limitation is that the present implementation remains a bounded laboratory proof of concept. The task setup, workspace, initial conditions, and fixtures are controlled. The ACT controller provides a deployable imitation-learning baseline, while language-conditioned checkpoint selection, unified multi-task VLA inference, autonomous planning, and generalized recovery remain outside the demonstrated scope.

6.2 Conclusions

The thesis presents an end-to-end engineering process for VLA-oriented bimanual industrial manipulation. Starting from customer requirements, the project defined measurable engineering specifications, generated modular solution concepts, selected a dual-FR3 and GELLO-based architecture, and implemented a data-driven manipulation pipeline using synchronized sensing and ACT policy deployment.

The completed system demonstrates that a dual-arm robot workcell can be integrated with teleoperation, dataset generation, learning-based control, and real-robot execution for the scoped bimanual manipulation task. The contribution is therefore both a working pro-

totype and a traceable design methodology: the final design choices can be related back to the QFD requirements, while the validation structure provides a basis for reporting task success, perception quality, pose accuracy, failure handling, and recovery once the detailed data section is completed.

Overall, the project establishes a practical foundation for flexible 3C assembly research. It shows that bimanual manipulation can be achieved within a controlled laboratory workcell using the selected hardware and learning pipeline, while leaving task-specific details and quantitative key data to be documented separately.

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Documentation of Generated Concepts Appendix 1

Purpose and design evolution

This appendix preserves the alternatives generated during the DR3 concept-development process and records the implementation-stage change from an ACT baseline to the LeRobot $\pi_{0.5}$ policy. The alternatives were retained because they expose the principal engineering trade-offs in platform selection, teleoperation, sensing, end-effector design, and learned control. The update to $\pi_{0.5}$ does not invalidate the earlier concept generation; it documents how implementation evidence and the requirement for language-conditioned multi-task control changed the final policy decision.

Table 1–1 Evolution from the DR3 baseline to the final implementation

Design aspect	DR3 baseline	Final implementation
Robot platform	Dual FR3 selected; single-arm evidence used for staged risk reduction	Dual FR3 physical execution
Policy	ACT in LeRobot; task-specific checkpoint selection	$\pi_{0.5}$ in LeRobot with language-conditioned task input
Task scope	Ring placement and plug insertion used as staged cases	Ring over can, plug insertion, and memory-module removal
Language	Future extension	Runtime task instruction supplied to the VLA policy
Control scope	Vision–action imitation-learning baseline	Vision–language–action bimanual control

Module alternatives retained after concept generation

C1: Robot-platform alternatives

The selected platform is the dual-FR3 workcell because it provides a direct controller path within the available laboratory ecosystem and supports coordinated bimanual manipulation. A dual Flexiv platform remains attractive for force-aware contact tasks, but it would require a different controller, teleoperation interface, and data-integration path. An ALOHA-

style platform offers a lower-cost bimanual research route, but its mechanical capability and software ecosystem are less directly aligned with the existing FR3 workcell. The selected platform accepts higher integration effort in exchange for continuity with the available hardware and a direct route to the final dual-arm system.

C2: Teleoperation alternatives

GELLO was selected because its leader-arm structure provides an intuitive relationship between operator motion and the robot state/action trajectories recorded for learning^[18]. PICO 4 Ultra offers an XR-controller and hand-tracking route, but would add motion-mapping and calibration work^[30]. The Universal Manipulation Interface (UMI) provides a portable hand-held demonstration approach^[19]; it remains a useful research comparator but is less directly coupled to the fixed dual-FR3 joint-state pipeline.

C3: Data-collection alternatives

The selected concept uses synchronized multi-view RGB observations together with the state and action channels of both arms and grippers. The global view provides workspace and fixture context, while wrist or side views preserve information around grasping and contact regions. A reduced one-wrist-plus-global configuration lowers storage, calibration, and synchronization burden, whereas a proprioception-only configuration removes visual synchronization entirely. Both alternatives reduce the information available for language-grounded object selection, occlusion handling, and contact-region observation.

C4: End-effector alternatives

The standard Franka parallel gripper provides a direct robot interface and repeatable parallel-jaw motion, but offers limited passive adaptation to object geometry. A rigid 3D-printed gripper can be tailored to one component and can provide accurate locating features. The compliant finger concept retained from DR3 offers greater tolerance to small geometric and pose variation and can reduce concentrated contact forces during grasping and insertion. Because the final system contains three materially different tasks, the deployed end-effector configuration must be documented per task rather than assuming that one finger geometry is optimal for the ring handle, plug body, and memory module.

C5: Policy alternatives

ACT was the deployable DR3 baseline because it was integrated with the existing LeRobot demonstration workflow and could predict temporally structured actions from images and robot state^[12]. Diffusion Policy remains a credible visuomotor baseline for contact-rich manipulation^[31]. Neither model natively provides the same language-conditioned task interface required by the final three-task system.

The final implementation therefore uses $\pi_{0.5}$ through LeRobot^[21,23]. Compared with the ACT baseline, the selected VLA policy accepts a natural-language task description in addition to visual and proprioceptive observations. This supports a common interface for the three demonstrated tasks and aligns the deployed controller with the original VLA objective. The selection is an implementation decision; superiority over ACT or Diffusion Policy should only be claimed if controlled comparative trials are reported.

Generated task concepts

The final demonstrations instantiate three task concepts from the common bimanual VLA architecture:

1. workspace preparation followed by ring grasping and placement over a can;
2. workspace preparation followed by grasping, alignment, and plug insertion;
3. motherboard repositioning followed by memory-module removal and placement in a tray.

These tasks were retained because they collectively exercise object repositioning, language-grounded object selection, bimanual coordination, grasping, transport, constrained insertion, component extraction, and terminal placement. Their exact language instructions and completion conditions are listed in Table 4–6.

Bill of materials Appendix 2

Table 2-1 Bill of materials

Quantity	Part Description	Purchased From	Part Number	Price (each)	Notes
...	...	...	...	...	...

Research Projects and Publications during Undergraduate Period

[1]

Acknowledgements

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A VISION-LANGUAGE-ACTION MODEL FOR DUAL-ARM ROBOTS IN 3C INDUSTRIAL PRODUCTION LINES

HCCI (Homogenous Charge Compression Ignition) combustion has advantages in terms of efficiency and reduced emission. HCCI combustion can not only ensure both the high economic and dynamic quality of the engine, but also efficiently reduce the NO_x and smoke emission. Moreover, one of the remarkable characteristics of HCCI combustion is that the ignition and combustion process are controlled by the chemical kinetics, so the HCCI ignition time can vary significantly with the changes of engine configuration parameters and operating conditions.